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Printing apparatus capable of improving convenience of setting sheet settings, method of controlling printing apparatus, and storage medium

Abstract

A printing apparatus executes a first notification using a first notification screen that, due to a difference of sheet information set in a received print job from the sheet information associated with the sheet setting applied to the printing apparatus when the print job has been received, prompts the user to change sheet information, and executes a second notification using a second notification screen, different from the first notification screen, that prompts, due to the difference, the user to confirm the sheet setting applied to the printing apparatus.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a printing apparatus, a method of controlling the printing apparatus, and a storage medium, and more particularly to a printing apparatus that has a print function capable of performing printing with a plurality of sheet settings, and is capable of improving the convenience of setting sheet settings, a method of controlling the printing apparatus, and a storage medium.

Description of the Related Art

(2) Conventionally, there is known a printing apparatus that is capable of using a variety of sizes and types of sheets as sheets used for printing. In a case where a printing apparatus of this type performs printing, such as copying, on a standalone basis, means is required which enables a user to select a sheet size and a type of sheet to be used (hereinafter such a combination of settings is referred to as “sheet settings”). Conventionally, for an inexpensive printing apparatus and a compact printing apparatus, there has been proposed, as such means, a simple display on which sheet settings can be selected. For example, Japanese Laid-Open Patent Publication (Kokai) No. 2019-4393 discloses a technique in which a user selects sheet settings using a main unit's user interface (UI) including a simple display that displays sheet settings by illuminating frames having respective predetermined shapes for number and character information with LED lights, and hard keys including a sheet selection button and a finalize button. In this technique, sheet sizes of predetermined types are sequentially switched and displayed on the display each time the sheet selection button is pressed, and when the finalize button is pressed, one of the sheet sizes displayed

on the display is finally determined as the size of sheets to be used when executing printing.

(3) Incidentally, as a method of setting sheet settings of the printing apparatus by operations performed on the printing apparatus comes into widespread use, there is an increasing demand for the improvement of the convenience of the printing apparatus using the method.

SUMMARY OF THE INVENTION

(4) The present invention provides a printing apparatus that is capable of improving the convenience of the printing apparatus using a method of setting sheet settings of the printing apparatus by operations performed on the printing apparatus, a method of controlling the printing apparatus, and a storage medium.

(5) In a first aspect of the present invention, there is provided a printing apparatus to which is applied a sheet setting selected by a user out of a plurality of sheet settings including a first sheet setting and a second sheet setting, wherein the first sheet setting is a setting with which is associated sheet information on a first sheet determined in advance without user's selection, wherein the second sheet setting is a setting with which is associated sheet information on a second sheet selected by the user out of the plurality of sheets, the printing apparatus including a reception unit configured to receive a print job, a print unit configured to execute printing based on the print job, and an execution unit configured to execute a first process before the printing based on the received print job is executed, based on occurrence of a first error that sheet information set in the received print job is different from sheet information associated with a sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and also the sheet information set in the received print job is not the first sheet information, and execute a second process different from the first process before the printing based on the received print job is executed, based on occurrence of a second error that the sheet information set in the received print job is different from the sheet information associated with the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, but the sheet information set in the received print job is the first sheet information.

(6) In a second aspect of the present invention, there is provided a method of controlling a printing apparatus to which is applied a sheet setting selected by a user out of a plurality of sheet settings including a first sheet setting and a second sheet setting, wherein the first sheet setting is a setting with which is associated sheet information on a first sheet determined in advance without user's selection, wherein the second sheet setting is a setting with which is associated sheet information on a second sheet selected by the user out of the plurality of sheets, the method including receiving a print job, executing printing based on the print job, executing a first process before the printing based on the received print job is executed, based on occurrence of a first error that sheet information set in the received print job is different from sheet information associated with a sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and also the sheet information set in the received print job is not the first sheet information, and executing a second process different from the first process before the printing based on the received print job is executed, based on occurrence of a second error that the sheet information set in the received print job is different from the sheet information associated with the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, but the sheet information set in the received print job is the first sheet information.

(7) According to the present invention, it is possible to improve the convenience of the printing apparatus using the method of setting sheet settings of the printing apparatus by operations performed on the printing apparatus.

(8) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing the whole configuration of a system including a printer as a printing apparatus according to a first embodiment of the present invention.
- (2) FIG. 2A is an appearance perspective view of the printer.
- (3) FIG. 2B is a schematic appearance view of a top surface of the printer.
- (4) FIG. 3 is a view showing the configuration of an operation panel as a main unit's user interface of the printer.
- (5) FIGS. 4A to 4C are diagrams each showing a sheet information-setting screen of a remote user interface (UI).
- (6) FIG. 5 is a block diagram showing a hardware configuration of the printer.
- (7) FIG. 6 is a diagram showing a module configuration of printer control software operating on the printer.
- (8) FIG. 7 is a block diagram showing a hardware configuration of a smartphone appearing in FIG. 1.
- (9) FIG. 8 is a diagram showing a module configuration of software operating on the smartphone.
- (10) FIG. 9 is a diagram showing a data structure of sheet information settings of the printer.
- (11) FIG. 10 is a flowchart of an event process performed by the printer.
- (12) FIG. 11 is a flowchart of a copy operation execution process in a step in FIG. 10.
- (13) FIG. 12 is a flowchart of a sheet selection operation process in a step in FIG. 10.
- (14) FIG. 13 is a flowchart of a sheet information-setting process in a step in FIG. 10.
- (15) FIG. 14 is a diagram showing a command list used in a driver job generated by a custom print driver of the smartphone.
- (16) FIG. 15 is a flowchart of a driver job execution process in a step in FIG. 10.
- (17) FIG. 16 is a diagram showing a data structure of a standard print job generated by a standard print service of the smartphone.
- (18) FIG. 17 is a flowchart of a standard job execution process in a step in FIG. 10.
- (19) FIG. 18 is a flowchart of a sheet mismatch error-handling process in a step in FIG. 15 and a step in FIG. 17, according to a second embodiment of the present invention.
- (20) FIGS. 19A and 19B are diagrams each showing an example of a screen displayed on the smartphone.

DESCRIPTION OF THE EMBODIMENTS

- (21) The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof. Note that the component elements described in the following embodiment are described only by way of example, and are by no means intended to limit the scope of the present invention to them alone.
- (22) FIG. 1 is a diagram showing the whole configuration of a system **100** including a printer **300** as a printing apparatus according to a first embodiment of the present invention.
- (23) Referring to FIG. 1, the system **100** is comprised of a cloud server **200** and a device group **400**, which are connected by a local area network (LAN) **102** and the Internet **104**.
- (24) The device group **400** includes a variety of apparatuses connectable via a network. In the present embodiment, as shown in FIG. 1, a smartphone **500**, the printer **300**, a client terminal **401**, such as a personal computer or a work station, and a digital camera **402** are included in the device group **400**, by way of example. However, types of the apparatuses included in the device group **400** are not limited to these but may include electrical home appliances, such as a refrigerator, a television, and an air conditioner.
- (25) The apparatuses included in the device group **400** are interconnected via the LAN **102** and can be connected to the Internet **104** via a router **103** connected to the LAN **102**. Here, although the

router **103** is illustrated as a device connecting between the LAN **102** and the Internet **104**, the router **103** can be equipped with a function of a wireless LAN access point (not shown) as a component of the LAN **102**. In this case, the device group **400** can not only connect to the router **103** by a wired LAN, but also connect to the access point by a wireless LAN to thereby participate in the LAN **102**. For example, it is possible to connect the printer **300** and the client terminal **401** to the router **103** by the wired LAN, and connect the smartphone **500** and the digital camera **402** to the router **103** (access point) by the wireless LAN. The device group **400** and the cloud server **200** can communicate with each other via the Internet **104** connected via the router **103**. Further, the variety of apparatuses included in the device group **400** can communicate with one another via the LAN **102**.

(26) Further, the smartphone **500** and the printer **300** can communicate with each other via short-range wireless communication **101**. The method of the short-range wireless communication **101** is not particularly limited, but for example, a method using wireless communication conforming to the Bluetooth (registered trademark) standard or the near field communication (NFC) standard can be employed.

(27) Further, the smartphone **500** is also connected to a mobile phone network **105** connected to the Internet **104** and can also communicate with the cloud server **200** via the mobile phone network **105**.

(28) Note that the configuration of the system **100**, shown in FIG. **1**, shows an example of the present invention, and there would be no change in the advantageous effects provided by the present invention even when a system having a different configuration is employed. For example, although the router **103** is equipped with the access point function by way of example, the access point used by the device group **400** to participate in the LAN **102** using the wireless LAN may be formed by a device different from the router **103**.

(29) FIGS. **2A** and **2B** are an appearance perspective view of the printer and a schematic appearance view of a top surface of the printer, respectively.

(30) As shown in FIGS. **2A** and **2B**, the printer **300** is a multifunction printer (MFP) equipped with a scanner function and other functions. Note that the printer **300** shows an example of the printing apparatus according to the present invention, and the printing apparatus is not limited to this insofar as it is an apparatus equipped with an operation panel **305** (main unit's user interface (UI)), described hereinafter, a print function capable of performing printing with a plurality of sheet settings, and a communication function of communicating with external terminal apparatuses including the smartphone **500**.

(31) FIG. **2A** is the appearance perspective view of the printer **300**.

(32) Referring to FIG. **2A**, the printer **300** includes an original platen glass **301**, an original platen glass pressing plate **302**, a print sheet insertion inlet **303**, and a printed sheet discharge outlet **304**.

(33) The original platen glass **301** is a transparent glass table and is used when an original image is read using a scanner mechanism after setting an original thereon. The original platen glass pressing plate **302** is a cover for pressing an original against the original platen glass **301** so as to prevent the original from floating when the original image is read by the scanner mechanism, and preventing outside light from entering the scanner mechanism of the printer **300**.

(34) The print sheet insertion inlet **303** is an insertion inlet for setting a variety of sizes of sheets. The sheets set on the print sheet insertion inlet **303** are conveyed into a print mechanism (not shown) arranged inside the printer **300** one by one, and discharged from the printed sheet discharge outlet **304** after being subjected to predetermined printing.

(35) FIG. **2B** is the schematic appearance view of the top surface of the printer **300**.

(36) Referring to FIG. **2B**, on the top of the original platen glass pressing plate **302**, there are arranged the operation panel **305**, a short-range wireless communication unit **306**, and a wireless LAN antenna **307**.

(37) The short-range wireless communication unit **306** is a unit for performing the short-range

wireless communication **101** and is capable of communicating with a short-range wireless communication unit of a communication partner (the smartphone **500** in the present embodiment) existing within a predetermined distance. The wireless LAN antenna **307** has an antenna embedded therein, for connecting to the access point described with reference to FIG. **1**, using the wireless LAN so as to participate in the LAN **102**.

(38) FIG. **3** is a view showing the configuration of the operation panel **305** as the main unit's user interface of the printer **300**.

(39) Referring to FIG. **3**, the operation panel **305** is formed by a variety of display lamps each using a light emitting diode (LED) and a variety of keys for receiving user's operations. However, the main unit's user interface of the printing apparatus according to the present invention is only required to be configured such that a user can select between a plurality of print settings, but the configuration of the main unit's user interface is not limited to that of the operation panel **305** in the present embodiment. For example, the main unit's user interface may be configured to have a liquid crystal screen or an organic electroluminescence (EL) screen, on which graphic display can be displayed, and a touch panel function.

(40) The variety of keys of the operation panel **305** include a power key **330**, a sheet selection key **337**, an OK key **338**, a monochrome start key **339**, a color start key **340**, and a stop key **341**.

Further, the variety of display lamps of the operation panel **305** include a power lamp **331**, a number-of-copies display section **332**, sheet selection lamps **333** to **336**, and an error lamp **342**.

(41) The power key **330** is a key for performing power-on and power-off operations. When the power key **330** is pressed by a user in a power-off state, the printer **300** is shifted to a power-on state. Inversely, when the power key **330** is pressed by a user in the power-on state, the printer **300** is shifted to the power-off state. Further, when the power key **330** is pressed by a user during printing operation or other processing operations, the printer **300** is shifted to the power-off state after completing or stopping the processing operation being performed.

(42) The power lamp **331** is a lamp indicating a power state of the printer **300**. The power lamp **331** is on (lights) in the power-on state and is off (extinguishes) in the power-off state. The power lamp **331** may be configured to blink e.g. during printing operation or like other processing operations, during shift to the power-on state, and during shift to the power-off state.

(43) The number-of-copies display section **332** is an indicator for displaying the number of sheets to be printed, including sheets to be copied. The number-of-copies display section **332** is formed by seven segments of LEDs and can display numbers of 0 to 9 by causing each segment as a component to be turned on and off. Further, the four sheet selection lamps **333** to **336** are provided adjacent to the number-of-copies display section **332**.

(44) The sheet selection lamp **333** labelled with a sheet selection option of "A4" is on in a case where an A4-sheet is selected as the print sheet by a user. The sheet selection lamp **334** labelled with a sheet selection option of "LTR" is on in a case where a letter sheet is selected as the print sheet by the user. The sheet selection lamp **335** labelled with a sheet selection option of "4×6" is on in a case where a 4×6 sheet for photo printing is selected as the print sheet by the user. The sheet selection lamp **336** labelled with a sheet selection option of "*" is on in a case where a sheet having a sheet size and a sheet type (sheet settings) registered for use by a method described hereinafter is selected as the print sheet by the user. That is, in the present embodiment, the printer **300** can use a set of sheet settings, which is associated with one sheet size (sheet type) predetermined without user's selection, out of sheet settings sets corresponding to the sheet selection lamps **333** to **335**, respectively. Further, in the present embodiment, the printer **300** can also use a set of sheet settings, which corresponds to the sheet selection lamp **336** and is associated with one sheet size selected by the user from a plurality of sheet sizes.

(45) Thus, the illustrated example in FIG. **3** takes the form in which the four sheet selection lamps are disposed at locations adjacent to the respective sets of sheet settings of "A4", "LTR", "4×6", and "*" which are physically printed or inscribed on the operation panel **305**. However, a form

different from this may be employed insofar as it can expressly display the sheet settings set currently selected by the user. For example, the operation panel **305** may be configured such that a lamp is incorporated in the notation itself of each sheet size, whereby a selected sheet is indicated by lighting on or off the notation. Further, the configuration may be such that a selected sheet is indicated using means other than an LED lamp.

(46) The sheet selection key **337** is for changing the sheet settings being currently set for the printer **300** by sequentially switching selection of the above-mentioned four sheet settings sets. A user sets a sheet settings set for the printer **300**, which is associated with sheets currently set for the printer **300**, by operating the sheet selection key **337**. This causes the printer **300** to perform printing on sheets corresponding to the sheet settings set currently set for the printer **300**. The OK key **338** is used to finally determine the sheet size for which one of the four sheet selection lamps is currently on (currently selected by the user). Besides for this, the OK key **338** is also sometimes used for another use according to the operating state of the printer **300**.

(47) The monochrome start key **339** and the color start key **340** are for starting a print operation, such an operation of the copy function, which is an operation performed by the printer **300** alone. When the monochrome start key **339** is pressed in a power-on and standby state, a monochrome copy operation is started in the printer **300**. Further, when the color start key **340** is pressed in the power-on and standby state, a color copy operation is started in the printer **300**. Further, the monochrome start key **339** and the color start key **340** are sometimes used to perform another operation or processing, according to an apparatus state of the printer **300** or a combination operation with another key.

(48) The stop key **341** is a key for interrupting an operation or processing being executed to return the printer **300** to the standby state. When the stop key **341** is pressed in an error state of the printer **300**, release of an error of the printer **300**, cancelation of a job being executed, or the like is performed.

(49) The error lamp **342** is a lamp which is off when the printer **300** is in the normal state and turns on or blinks when an error occurs in the printer **300**. Errors occurring in the printer **300** include a plurality of types, such as a no ink error, an out-of-paper error, and a paper jam error, and hence the error lamp **342** can notify a user of a type of the error currently occurring in the printer **300** by changing the blinking pattern. Further, by controlling the display of the number-of-copies display section **332** in combination the turn-on or blinking of the error lamp **342**, it is also possible to notify a user of a type of the error and an error handling method.

(50) FIGS. **4A** to **4C** are diagrams each showing a sheet information-setting screen **600** of a remote user interface.

(51) The remote user interface (display section) is a screen corresponding to an operation screen of the printer **300** and is a user interface displayed by the printer **300** on a display section of an external terminal apparatus, such as the smartphone **500** or the client terminal **401**. The printer **300** distributes, as a Web server, remote user interface contents stored therein to an external terminal apparatus, whereby the external terminal apparatus displays the remote user interface using a Web browser function and receives an operation from a user via the interface. Note that based on the received operation, the external terminal apparatus transmits an instruction based on the received operation to the printer **300** as required. With this, the printer **300** executes processing based on the operation received by the external terminal apparatus via the remote user interface. The processing based on the operation received by the external terminal apparatus via the remote user interface is, more specifically, e.g. processing for determining one of a plurality of sheet information items as the sheet information corresponding to the sheet selection lamp **336**, as will be described hereinafter.

(52) FIG. **4A** is a diagram showing the configuration of a main screen of the sheet information-setting screen **600**.

(53) The sheet information-setting screen **600** includes sheet information display sections **601** to

604 which correspond to the four sheet selection lamps **333** to **336** provided on the operation panel **305** of the printer **300**, and display four sheet information items, respectively, in a list, each formed by settings of the sheet size and the sheet type.

(54) On the sheet information display section **601**, “BODY PANEL SPECIFICATION: A4” as the sheet information corresponding to the sheet selection lamp **333** is displayed.

(55) On the sheet information display section **602**, “BODY PANEL SPECIFICATION: LTR” as the sheet information corresponding to the sheet selection lamp **334** is displayed.

(56) On the sheet information display section **603**, “BODY PANEL SPECIFICATION: 4×6” as the sheet information corresponding to the sheet selection lamp **335** is displayed.

(57) On the sheet information display section **604** (registered sheet information display section), “BODY PANEL SPECIFICATION: *” as the sheet information corresponding to the sheet selection lamp **336** is displayed. A sheet size-changing button **605** is provided adjacent to the display of the sheet size of the sheet information display section **604**. When a user presses the sheet size-changing button **605**, a sheet size selection screen **620** (see FIG. 4B) is displayed. The user can select a desired one of a plurality of sheet sizes **621** on the sheet size selection screen **620** to change the sheet size associated with the sheet selection lamp **336** to the selected sheet size. That is, the plurality of sheet sizes **621** are a plurality of sheet settings which can be set on the remote user interface as one of the sheet settings associated with the sheet selection lamp **336**. In other words, the plurality of sheet sizes **621** are candidates of one of the sheet settings set on the remote user interface as the one of the sheet settings associated with the sheet selection lamp **336**. On the lowest part of the sheet size selection screen **620**, there is provided an OK button **622**. When the OK button **622** is pressed, a sheet size selected on the sheet size selection screen **620** is set as a sheet size associated with the sheet selection lamp **336**, and the displayed screen returns to the sheet information-setting screen **600**. Similarly, a sheet type-changing button **606** is provided adjacent to the display of the sheet type of the sheet information display section **604**. When the user presses the sheet type-changing button **606**, a sheet type selection screen **630** (see FIG. 4C) is displayed, and the user can select a desired one of a plurality of sheet types on this screen to change the sheet type associated with the sheet selection lamp **336** to the selected sheet type. On the lowest part of the sheet type selection screen **630**, there is provided an OK button **632**. When the OK button **632** is pressed, a sheet type selected on the sheet size selection screen **620** is set as a sheet type associated with the sheet selection lamp **336**, and the displayed screen returns to the sheet information-setting screen **600**.

(58) On the lowest part of the sheet information-setting screen **600**, there are arranged an OK button **610** and a cancel button **611**. When the OK button **610** is pressed, the information on the sheet size and the sheet type of a registered sheet, whose settings have been changed by the user using the sheet size selection screen **620** (see FIG. 4B) and the sheet type selection screen **630** (see FIG. 4C), is reflected on the printer **300**, and the display of the sheet information-setting screen **600** is terminated. When the cancel button **611** is pressed, the information on the sheet size and the sheet type of the registered sheet, whose settings have been changed by the user, is discarded, and the display of the sheet information-setting screen **600** is terminated without reflecting on the printer **300** these information items which have been changed.

(59) FIG. 5 is a block diagram showing a hardware configuration of the printer **300**.

(60) Referring to FIG. 5, the printer **300** includes a main board **310** that controls the overall operation of the apparatus, a wireless LAN unit **308**, the short-range wireless communication unit **306**, and the operation panel **305**.

(61) The main board **310** includes a central processing unit (CPU) **311**, a program memory **313**, a data memory **314**, a scanner mechanism control circuit **315**, a print mechanism control circuit **317**, a wireless LAN control circuit **318**, a short-range wireless communication control circuit **319**, and an operation section control circuit **320**. These devices are interconnected via an internal bus **312**.

(62) The CPU **311** is a microprocessor that controls the devices in the main board **310**, and the

program memory **313** is a read only memory (ROM) and stores control programs. The data memory **314** is a random access memory (RAM) and includes an image memory **316**. The CPU **311** operates according to the control programs stored in the program memory **313** and a variety of data stored in the data memory **314**.

(63) The scanner mechanism control circuit **315** controls the scanner mechanism to read an image of an original under the control of the CPU **311**. The image of the original, read here, is stored in the image memory **316** included in the data memory **314**.

(64) The print mechanism control circuit **317** prints an image stored in the image memory **316** included in the data memory **314** on a recording medium under the control of the CPU **311**.

(65) The wireless LAN control circuit **318** controls, under the control of the CPU **311**, the wireless LAN unit **308** connected thereto, to connect to the above-described access point to participate in the LAN **102**, and thereby communicate with another communication terminal apparatus.

(66) The short-range wireless communication control circuit **319** controls, under the control of the CPU **311**, the short-range wireless communication unit **306** connected thereto, to detect connection of another short-range wireless communication terminal and transmit and receive data to and from the other short-range wireless communication terminal.

(67) The operation section control circuit **320** controls, under the control of the CPU **311**, the operation panel **305** connected thereto, to display a state of the printer **300**, display a function selection menu of the printer **300**, and receive an operation from a user.

(68) FIG. **6** is a diagram showing a module configuration of printer control software **1300** operating on the printer **300**.

(69) The printer control software **1300** operating on the printer **300** operates on a built-in control operating system (OS) **1301**. Further, the modules of the printer control software **1300** are roughly classified into a system control layer **1310**, a job management layer **1320**, a middleware layer **1330**, and an application layer **1340**.

(70) The built-in control OS **1301** is an OS that controls the basic operation of the printer control software **1300**, and a real-time OS excellent in responsiveness is generally used.

(71) The system control layer **1310** is formed by a control module group that mainly controls the hardware of the printer **300**. Here, the system control layer **1310** includes a print control module **1311**, a read control module **1312**, a panel control module **1313**, a memory management module **1314**, a power supply control module **1315**, a short-range wireless communication control module **1316**, and a network communication control module **1317**.

(72) The print control module **1311** is a control module for controlling the print mechanism control circuit **317** for execution of the print operation of the printer **300**. The read control module **1312** is a control module for controlling the scanner mechanism control circuit **315** for execution of the operation for reading an original placed on the original platen glass **301**. The panel control module **1313** is a control module for controlling the display on the operation panel **305** of the printer **300** and detecting a variety of key operations. The memory management module **1314** is a control module for performing management, such as dynamic allocation of a variety of data items stored in the data memory **314**. The power supply control module **1315** is a control module for performing power supply control for the printer **300**, supplying electric power necessary for operation to each hardware block, and performing power-saving mode control. The short-range wireless communication control module **1316** (acquisition unit) is a control module for controlling the short-range wireless communication unit **306** for execution of short-range wireless communication with a terminal apparatus, such as the smartphone **500**, so as to perform acquisition of a job and the like. The network communication control module **1317** (acquisition unit) is a control module for controlling the wireless LAN unit **308** for communication with an external terminal apparatus and thereby perform control of a communication physical layer so as to perform acquisition of a job and the like.

(73) The job management layer **1320** is formed by a module group for executing a variety of

operations by using the system control layer **1310**, while performing hardware resource allocation, exclusive control, scheduling, and so forth, in response to job execution requests from an upper layer.

(74) The middleware layer **1330** is positioned between the application layer **1340** and the job management layer **1320**, and is formed by a module group commonly used by a plurality of function modules included in the application layer **1340**. Here, the middleware layer **1330** includes an application framework **1331**, a network protocol stack **1332**, an encryption processing module **1333**, and a Web server module **1334**.

(75) The application framework **1331** is a framework module commonly used when job execution is requested from the application layer **1340** to the job management layer **1320**. The network protocol stack **1332** is a module for performing communication based on a variety of network protocols, such as HTTP and TCP/IP. The encryption processing module **1333** is a module for performing encryption and decryption processing necessary for network communication and the like. The Web server module **1334** is a module for causing the printer **300** to operate as the Web server to distribute the remote user interface contents.

(76) The application layer **1340** is formed by an application group for realizing a variety of functions of the printer **300**. Here, the application layer **1340** includes a copy application **1341**, a driver print application **1342**, a standard print application **1343**, and a remote user interface application **1344**.

(77) The copy application **1341** is an application module for executing a copy operation for reading an original and printing read data. The driver print application **1342** is an application module for receiving jobs from printer drivers of the smartphone **500** and the client terminal **401** and executing associated operations. The jobs received from these printer drivers include a print job for executing a print operation, a scan job for reading an original and outputting image data, a maintenance job for configuring and managing the printer **300** by exchanging information with an external terminal apparatus, and so forth. The standard print application **1343** is an application module for receiving jobs from a standard print service included in the system of the smartphone **500** and executing associated operations. The jobs received from the standard print service include a print job for executing a print operation and a scan job for reading an original and outputting image data. However, the standard print service is provided by a smartphone maker and a standard OS vendor, and hence there is restriction that it is impossible to perform specific setting and management unique to a printer model of each maker. The remote user interface application **1344** is a module for providing the remote user interface function equipped in the printer **300** to an external terminal apparatus using the Web server module **1334**. By using the remote user interface, the external terminal apparatus is enabled to perform setting and management of the printer **300**. Particularly, in the printer **300** described in the present embodiment, the information which can be displayed and set on the operation panel **305** is limited, and hence the remote user interface is used to perform more detailed setting and management. Further, in a case where a dedicated driver is not available in the smartphone **500** and the client terminal **401**, only the standard print service can be used, and hence the use of the remote user interface is essential to perform the detailed setting and management.

(78) FIG. 7 is a block diagram showing a hardware configuration of the smartphone **500**.

(79) Referring to FIG. 7, the smartphone **500** includes a main board **510** that controls the overall operation of the apparatus, a wireless LAN unit **502**, a short-range wireless communication unit **501**, a line connection unit **503**, and a touch panel display **504**.

(80) The main board **510** includes a CPU **511**, a program memory **513**, a data memory **514**, a wireless LAN control circuit **515**, a short-range wireless communication control circuit **516**, a line control circuit **517**, an operation section control circuit **518**, a camera **519**, and a nonvolatile memory **521**. These devices are interconnected via an internal bus **512**.

(81) The CPU **511** is a microprocessor that controls the devices in the main board **510**, and the

program memory **513** is a ROM and stores control programs. The data memory **514** is a RAM and includes an image memory **520**. The CPU **511** operates according to the control programs stored in the program memory **513** and a variety of data stored in the data memory **514**.

(82) The wireless LAN control circuit **515** controls, under the control of the CPU **511**, the wireless LAN unit **502** connected thereto, to connect to the above-described access point to participate in the LAN **102**, and thereby communicate with another communication terminal apparatus.

(83) The short-range wireless communication control circuit **516** controls, under the control of the CPU **511**, the short-range wireless communication unit **501** connected thereto, to detect connection of another short-range wireless communication terminal and transmit and receive data to and from the other short-range wireless communication terminal.

(84) The line control circuit **517** controls the line connection unit **503** connected thereto, to connect to the mobile phone network **105** to perform a call and data transmission/reception under the control of the CPU **511**.

(85) The operation section control circuit **518** performs, under the control of the CPU **511**, desired display on the touch panel display **504** connected thereto, and receives an operation from a user.

(86) The camera **519** photographs, under the control of the CPU **511**, an image and stores the photographed image in the image memory **520** included in the data memory **514**. Further, not only the image photographed by the camera **519**, but also images acquired from the outside via the mobile phone line, the LAN **102**, and the short-range wireless communication **101** are stored in the image memory **520**, and these images can be reversely transmitted from the image memory **520** to the outside.

(87) The nonvolatile memory **521** is implemented e.g., by a flash memory and stores data which is desired to be stored even after the power is turned off. The nonvolatile memory **521**, stores, for example, not only address book data, a variety of communication connection information items, information on devices connected in the past, and so forth, but also image data desired to be stored, application software for realizing a variety of functions on the smartphone **500**, and so forth.

(88) FIG. **8** is a diagram showing a module configuration of software **1500** operating on the smartphone **500**. Note that although in the present embodiment, the apparatus on which the software **1500** operates is the smartphone **500**, this is not limitative, but for example, the apparatus on which the software **1500** operates may be the client terminal **401** or the digital camera **402** insofar as it is an external terminal apparatus that can communicate with the printer **300**.

(89) The software **1500** operating on the smartphone **500** operates on an operating system (OS) **1501**. Further, the modules of the software **1500** are roughly classified into a system control layer **1510**, a middleware layer **1520**, and an application layer **1530**.

(90) The OS **1501** is an OS that controls the basic operation of the software **1500**.

(91) The system control layer **1510** is mainly formed by a module group that controls the hardware of the smartphone **500**. Here, the system control layer **1510** includes a graphical user interface (GUI) control module **1511**, a camera control module **1512**, a short-range wireless communication control module **1513**, a network communication control module **1514**, a memory management module **1515**, and a power supply control module **1516**.

(92) The GUI control module **1511** is a control module for controlling the display on the touch panel display **504** and detecting a user operation on the touch panel display **504**. The camera control module **1512** is a module for controlling photographing performed by the camera **519**. The short-range wireless communication control module **1513** is a control module for controlling the short-range wireless communication module **501** to perform short-range wireless communication with another apparatus, such as the printer **300**. The network communication control module **1514** is a control module for controlling the wireless LAN unit **502** to connect to the above-described access point to participate in the LAN **102**, and thereby perform communication physical layer control for performing communication with an external terminal apparatus. The memory management module **1515** is a control module for performing management, such as dynamic

allocation of a variety of data items stored in the data memory **514**. The power supply control module **1516** is a control module for performing power supply control for the smartphone **500** to supply electric power necessary for operation to each hardware block and perform power-saving mode control.

(93) The middleware layer **1520** is positioned between the application layer **1530** and the system control layer **1510** and is formed by a module group commonly used by a plurality of function modules included in the application layer **1530**. Here, the middleware layer **1520** includes an application framework **1521**, a standard print service **1522**, and a network protocol stack **1523**.

(94) The application framework **1521** is a framework module commonly used when execution of a job is requested from the application layer **1530** to the system control layer **1510**. The network protocol stack **1523** is a module for performing communication based on a variety of network protocols, such as HTTP and TCP/IP. The standard print service **1522** is a service module for generating a standard job to provide functions using the printer **300**, which are associated with the standard print service **1522**, to the applications of the application layer **1530**. The functions provided by the standard print service **1522** include a print function of causing the printer **300** to execute a standard print job generated as a standard job, a scan function of causing the printer **300** to execute a standard scan job generated as a standard job, and so forth. However, the standard print service is provided by a smartphone maker or a standard OS vendor, and hence there is restriction that it is impossible to perform specific setting and management unique to a printer model of each maker.

(95) The application layer **1530** is formed by an application group for realizing a variety of functions of the smartphone **500**, and the like. Here, the application layer **1530** includes a first application **1531**, a second application **1532**, a custom print driver **1533**, and a Web browser **1534**.

(96) The applications installed in the smartphone **500** include those incorporated at the time of product shipment and those which can be downloaded and introduced by a user afterwards. Although FIG. **8** shows the first application **1531** and the second application **1532** as the applications installed in the smartphone **500** by way of example, the number and types of applications installed in the smartphone **500** are not limited, and the user can add and delete applications as required. The custom print driver **1533** is a dedicated driver prepared for the model type of the printer **300** and provides functions for executing a print job and a scan job to the associated printer **300**. The user selects and installs a print driver associated with the model type of the printer **300** used by himself/herself as the custom print driver **1533**. Note that although the custom print driver **1533** is illustrated here as installed in the application layer **1530** by way of example, the custom print driver **1533** may be configured to be installed in the middleware layer **1520** and operate according to a print job request received from an application. The Web browser **1534** is an application module for connecting to a Web server via the LAN **102** to acquire Web contents and display the acquired Web contents. The Web browser **1534** can acquire Web contents from the cloud server **200** and display the acquired Web contents, and acquire and display remote user interface contents distributed from the printer **300**.

(97) FIG. **9** is a diagram showing a data structure of sheet information settings **5000** of the printer **300**.

(98) The sheet information settings **5000** is formed by sheet settings selection values **5001** of 1 to 4 and sheet information data sets **5002** to **5005** corresponding to the four types of sheet information items, respectively, described with reference to FIGS. **4A** to **4C**.

(99) The sheet information data sets **5002** to **5005** have sheet size fields **5006**, **5008**, **5010**, and **5012** each for storing a sheet size, and sheet type fields **5007**, **5009**, **5011**, and **5013** each for storing a sheet type, respectively. Further, values of 1 to 4, which correspond to the sheet settings selection values **5001**, respectively, are assigned to the sheet information data sets **5002** to **5005**, as respective IDs.

(100) IDs for identifying respective sheet sizes are stored in the sheet size fields **5006**, **5008**, **5010**,

and **5012**, respectively. Here, the sheet sizes listed on the sheet size selection screen **620** shown in FIG. **4B** have values of 1 to 14 sequentially assigned thereto from the top of the list, as respective IDs. For example, the sheet size set for the sheet information data set **5002** (specified sheet option) is A4 which is a sheet size of the sheet settings specified in advance, and hence the value of an ID stored in the sheet size field **5006** is fixed to the default value of “1”. Further, the sheet sizes set for the sheet information data sets **5003** and **5004** are LTR and 4×6 which are sheet sizes of the sheet settings specified in advance. Therefore, the values of IDs stored in the sheet size fields **5008** and **5010** are fixed to the default values of “10” and “8”, respectively. Note that in the sheet size field **5012** in the sheet information data set **5005** (registered sheet option), a value of 0 is stored in a case where no sheet size is set, and in a case where a sheet size is set by a user, a desired one of the ID values of 1 to 14, for identifying the set sheet size, is stored.

(101) Further, in the sheet type fields **5007**, **5009**, **5011**, and **5013**, IDs for identifying sheet types are stored, respectively. Here, the sheet types listed on the sheet type selection screen **630** shown in FIG. **4C** have values of 1 to 5 sequentially assigned thereto from the top of the list, as respective IDs. For example, the sheet type set for the sheet information data set **5002** is plain paper which is a sheet type of the sheet settings specified in advance, and hence the value of an ID stored in the sheet type field **5007** is fixed to the default value of “1”. Similarly, the sheet types set for the sheet information data sets **5003** and **5004** are plain paper and glossy paper which are sheet types of the sheet settings specified in advance, and hence the values of IDs stored in the sheet type fields **5009** and **5011** are fixed to the default values of “1” and “2”, respectively. Note that in the sheet type field **5013** in the sheet information data set **5005**, a value of 0 is stored in a case where no sheet type is set, and in a case where a sheet type is set by a user, a desired one of the ID values of 1 to 5, for identifying the set sheet type, is stored.

(102) When the printer **300** is operated, a sheet information data set having an ID corresponding to a selected one of the sheet settings selection values **5001** is referred to, whereby the sheet size and the sheet type to be used are determined. For example, in a case where the sheet settings selection value **5001** of “1” is selected, the sheet information data set **5002** is referred to, and the sheet size “A4” and the sheet type “plain paper” are determined as the sheet size and the sheet type to be used when the printer **300** is operated.

(103) The sheet information settings **5000** having the data structure shown in FIG. **9** is stored in the data memory **314** and initialized at the start of the printer **300** for use. The set ID values are stored in the nonvolatile memory from which the data is not deleted even after the power supply to the printer **300** is cut off.

(104) FIG. **10** is a flowchart of an event process performed by the printer **300**. When the AC power is supplied to the printer **300**, the CPU **311** of the printer **300** starts execution of the event process by reading a program thereof from the program memory **313**.

(105) First, in a step **S2001**, the related hardware components and the data memory **314** of the printer **300** are initialized. In this step, the printer **300** is shifted to the power-off state waiting for a power-on operation, and the process proceeds to a step **S2002**.

(106) In the step **S2002**, the CPU **311** waits for occurrence of an event in the power-off state, and when an event occurs, the process proceeds to a step **S2003**. Examples of the event whose occurrence is monitored for here include a user's operation of pressing the power key **330**. In the present process, a different branching process is performed according to an occurred event.

(107) In the step **S2003**, if it is detected that the occurred event is the operation of pressing the power key **330** (YES to the step **S2003**), the process proceeds to a step **S2004**, wherein the printer **300** is shifted to the power-on state, and the process proceeds to a step **S2005**. Otherwise (NO to the step **S2003**), the process returns to the step **S2002**, wherein the CPU **311** waits for occurrence of the next event.

(108) In the step **S2005**, the CPU **311** waits for occurrence of an event in the power-on state, and when an event occurs, the process proceeds to a step **S2006**. Examples of the event whose

occurrence is monitored for here include a user's operation of pressing the power key **330** and a key operation on the operation panel **305**. Further, reception of job data from an external terminal apparatus via the short-range wireless communication, a remote user interface access request from an external terminal apparatus to the Web server module **1334** of the printer **300**, and so forth are also included in the examples of the event. In the present process, a different branching process is performed according to an occurred event.

(109) In the step **S2006**, if it is detected that the occurred event is the operation of pressing the power key **330** (YES to the step **S2006**), the process proceeds to a step **S2007**, wherein the printer **300** is shifted to the power-off state, and then the process returns to the step **S2002**. Otherwise (NO to the step **S2006**), the process proceeds to a step **S2008**.

(110) In the step **S2008**, if it is detected that the occurred event is an operation of pressing the color start key **340** (YES to the step **S2008**), the process proceeds to a step **S2009**, wherein a copy operation execution process (color), described hereinafter, is executed, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2008**), the process proceeds to a step **S2010**.

(111) In the step **S2010**, if it is detected that the occurred event is an operation of pressing the monochrome start key **339** (YES to the step **S2010**), the process proceeds to a step **S2011**, wherein a copy operation execution process (monochrome), described hereinafter, is executed, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2010**), the process proceeds to a step **S2012**.

(112) In the step **S2012**, if it is detected that the occurred event is reception of a driver job from the custom print driver **1533** of the smartphone **500** (YES to the step **S2012**), the process proceeds to a step **S2013**, wherein a driver job execution process is executed, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2012**), the process proceeds to a step **S2014**. Note that details of the driver job execution process will be described hereinafter.

(113) In the step **S2014**, if it is detected that the occurred event is reception of a standard job from the standard print service **1522** of the smartphone **500** (YES to the step **S2014**), the process proceeds to a step **S2015**, wherein a standard job execution process, described hereinafter, is executed, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2014**), the process proceeds to a step **S2016**.

(114) In the step **S2016**, if it is detected that the occurred event is an operation of pressing the sheet selection key **337** (YES to the step **S2016**), the process proceeds to a step **S2017**, wherein a sheet selection operation process, described hereinafter, is executed, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2016**), the process proceeds to a step **S2018**.

(115) In the step **S2018**, if it is detected that the occurred event is an operation of pressing the OK key **338** (YES to the step **S2018**), the process proceeds to a step **S2019**, wherein the number (the number of copies) displayed on the number-of-copies display section **332** is incremented by 1, and then the process returns to the step **S2005**. This number of copies is used when the copy operation is executed. Otherwise (NO to the step **S2018**), the process proceeds to a step **S2020**.

(116) In the step **S2020**, if it is detected that the occurred event is an operation of pressing the stop key **341** (YES to the step **S2020**), the process proceeds to a step **S2021**, wherein the number (the number of copies) displayed on the number-of-copies display section **332** is cleared and set to the initial value of 1, and then the process returns to the step **S2005**. Otherwise (NO to the step **S2020**), the process proceeds to a step **S2022**.

(117) In the step **S2022**, if it is detected that the occurred event is a remote user interface access request from the Web browser **1534** of the smartphone **500** to the Web server module **1334** of the printer **300** (YES to the step **S2022**), the process proceeds to a step **S2023**. Otherwise (NO to the step **S2022**), the process directly returns to the step **S2005**.

(118) In the step **S2023**, a sheet information-setting process is executed, and then the process returns to the step **S2005**. Note that details of the sheet information-setting process will be described hereinafter.

(119) FIG. 11 is a flowchart of the copy operation execution process in the step S2009 and the step S2011 in FIG. 10. Note that although color copy is executed in the copy operation execution process in the step S2009, and monochrome copy is executed in the copy operation execution process in the step S2011, these processes are different only in that the scan and print operations are executed for color or for monochrome, but there is no difference in the flowchart in FIG. 11. In the present process, the sheet information data set having an ID corresponding to a selected one of the sheet settings selection values 5001 is referred to, whereby the sheet size and the sheet type to be used are determined. These settings are used for the size of originals to be read in a scan operation, and the sheet size and the sheet type in a print operation.

(120) First, in a step S2101, the scanner mechanism in the printer 300 is operated to start the operation of reading an original placed on the original platen glass 301.

(121) In the next step S2102, the print mechanism is operated to start the operation for printing one page of the read image. Note that the reading operation and the printing operation are executed as different processes operating on the real-time OS and executed in parallel with the present process. The CPU 311 executes the following process while monitoring the status of execution of these different processes.

(122) In a step S2103, the CPU 311 checks whether or not the operation for printing one page (page print) is completed. If the operation for printing one page is completed (YES to the step S2103), the process proceeds to a step S2104, wherein the number (the number of copies) displayed on the number-of-copies display section 332 is decremented by 1, and the process proceeds to a step S2105.

(123) In the step S2105, the CPU 311 checks whether or not the number of copies is equal to 0, i.e. whether or not printing of the set number of copies is completed. If the number of copies is equal to 0 (YES to the step S2105), the present process is terminated, whereas if the number of copies is not equal to 0 (NO to the step S2105), the process returns to the step S2102, and the operation for printing the next page is started.

(124) On the other hand, if it is determined in the step S2103 that completion of the operation for printing one page is being continuously monitored (NO to the step S2103), the process proceeds to a step S2106 to check whether or not one of the plurality of events has occurred.

(125) In the step S2106, if the event of pressing the stop key 341 has been detected (YES to the step S2106), the process proceeds to a step S2107, wherein the print operation being executed is stopped, followed by terminating the present process. On the other hand, if the event of pressing the stop key 341 has not been detected (NO to the step S2106), the process proceeds to a step S2108.

(126) In the step S2108, if the event of pressing the monochrome start key 339 or the color start key 340 has been detected (YES to the step S2108), the process proceeds to a step S2109, wherein the value of the number-of-copies display section 332 is incremented by 1, and then the process returns to the step S2103. Otherwise (NO to the step S2108), the process directly returns to the step S2103.

(127) FIG. 12 is a flowchart of the sheet selection operation process in the step S2017 in FIG. 10.

(128) First, in a step S2201, the CPU 311 acquires the default value of the sheet settings selection values 5001 and stores the acquired value as a variable "x" in the work area in the data memory 314.

(129) In the next step S2202, a lamp blinking operation is started in which one of the sheet selection lamps 333 to 336, which corresponds to the sheet settings selection value 5001 indicated by the variable "x", is blinked. Note that this lamp blinking operation is executed as another process operating on the real-time OS in parallel with the present process.

(130) In the next step S2203, the CPU 311 waits for occurrence of an event, and when an event occurs, the process proceeds to a step S2204. The event whose occurrence is monitored for here is a key operation performed by the user on the operation panel 305. In the present process, a

different branching process is performed according to an occurred event.

(131) If it is detected in the step **S2204** that the occurred event is the operation of pressing the sheet selection key **337** (YES to the step **S2204**), steps **S2205** to **S2207** are repeated to perform processing for sequentially switching the sheet selection values. More specifically, whenever the sheet selection key **337** is pressed, the process proceeds to a step **S2205**, wherein the variable “x” is incremented by 1. In the next step **S2206**, the CPU **311** checks whether or not the value of the variable “x” is larger than 4. As a result, if the value of the variable “x” is larger than 4 (YES to the step **S2206**), the process proceeds to a step **S2207**, wherein the value of 1 is substituted for the variable “x”, and then the process returns to the step **S2202** to blink one of the sheet selection lamps **333** to **336**, which corresponds to the value of the updated variable, i.e. 1. Otherwise (NO to the step **S2206**), the process directly returns to the step **S2202** to blink one of the sheet selection lamps **333** to **336**, which corresponds to the updated variable “x”.

(132) On the other hand, if the occurred event is not the operation of pressing the sheet selection key **337** (NO to the step **S2204**), the process proceeds to a step **S2208**.

(133) If it is detected in the step **S2208** that the occurred event is the operation of pressing the OK key **338** (YES to the step **S2208**), steps **S2209** and **S2210** are executed, and the sheet selection performed by the user is finalized. More specifically, when the OK key **338** is pressed, the process proceeds to the step **S2209**, wherein the value of the variable “x” is substituted for the sheet settings selection value **5001**. Then, in the step **S2210**, one of the sheet selection lamps **333** to **336**, which correspond to the sheet settings selection value **5001** thus set, is turned on, followed by terminating the present process.

(134) On the other hand, if the occurred event is not the operation of pressing the OK key **338** (NO to the step **S2208**), the process proceeds to a step **S2211**.

(135) If it is detected in the step **S2211** that the occurred event is the operation of pressing the stop key **341** (YES to the step **S2211**), the CPU **311** executes the step **S2210** to perform processing for stopping the sheet selection operation performed by the user. More specifically, when the stop key **341** is pressed, the process proceeds to the step **S2210**, wherein one of the sheet selection lamps **333** to **336**, which corresponds to the sheet settings selection value **500**, is turned on, followed by terminating the present process. At this time, the present process is terminated without substituting the value of the variable “x” for the sheet settings selection value **5001**, and hence the state of the printer **300** returns to that before starting the sheet selection operation.

(136) FIG. **13** is a flowchart of the sheet information-setting process in the step **S2023** in FIG. **10**. As described above, the present process is executed in a case where a remote user interface access request from the Web browser **1534** of the smartphone **500** to the Web server module **1334** of the printer **300** is detected as the occurred event.

(137) First, in a step **S2301**, a response to the remote user interface access request is transmitted to the Web browser **1534** of the smartphone **500**. The contents of the response are those of the sheet information-setting screen **600** illustrated in FIG. **4A**.

(138) In a step **S2302**, the CPU **311** waits for occurrence of an event, and when an event occurs, the process proceeds to a step **S2303**. Examples of the event whose occurrence is monitored for here include a request from the smartphone **500**, closing of a session or disconnection, and time-out. In the present process, a different branching process is performed according to an occurred event.

(139) If it is detected in the step **S2303** that the occurred event is a POST request from the smartphone **500** to the remote user interface (operation of a detection unit) (YES to the step **S2303**), the process proceeds to a step **S2304**.

(140) In the step **S2304**, the CPU **311** updates the sheet information data set **5005** (see FIG. **9**) which is the sheet information which can be registered by the user (operation of a registration unit). More specifically, the sheet size and the sheet type, selected by the user in the sheet information display section **604** on the sheet information-setting screen **600** (see FIG. **4A**) of the remote user interface are described in the POST request detected in the step **S2303**. The sheet information data

set **5005** appearing in FIG. **9** is updated based on the sheet size and the sheet type, described in this POST request.

(141) On the other hand, if the occurred event is not a POST request from the smartphone **500** to the remote user interface (NO to the step **S2303**), the process proceeds to a step **S2305**.

(142) If it is detected in the step **S2305** that the occurred event is closing of a session or disconnection (YES to the step **S2305**), the present process is terminated, whereas if not (NO to the step **S2305**), the process proceeds to a step **S2306**.

(143) If it is detected in the step **S2306** that the occurred event is time-out (YES to the step **S2306**), the present process is similarly terminated, whereas if not (NO to the step **S2306**), the process returns to the step **S2302**. Note that the event of time-out is an event caused by processing of a timer incorporated in the printer **300** in a case where communication from the smartphone **500** has not been performed for a predetermined time period or more.

(144) FIG. **14** is a diagram showing a command list **5100** used in a driver job generated by the custom print driver **1533** of the smartphone **500**.

(145) The driver job is formed by a command line which is generated based on one of predetermined command specifications in command list **5100** shown in FIG. **14**. In the printer **300**, the commands of the driver job from the smartphone **500**, reception of which has been detected in the step **S2012**, are sequentially analyzed by the driver print application **1342**, and processing operations corresponding to the analyzed commands are executed. Each command transmitted from the custom print driver **1533** to the printer **300** is comprised of a command number **5101**, a data length **5102**, and data **5103** of bytes in a number indicated by the data length.

(146) The commands shown in FIG. **14**, which are used in driver jobs, will be described in detail hereafter.

(147) A job start command is a command to declare the start of a driver job and can pass the type of the driver job as data. A job end command is a command to declare the end of the driver job. A job cancel command is a command to stop operation during execution of the driver job and cancel the driver job.

(148) A print job command group includes a print parameter setting command, a page start command, a print data transfer command, a page end command, and so forth.

(149) The print parameter setting command is a command to designate a sheet size and a sheet type of sheets used in the print job and a print quality. The sheet size and the sheet type, designated in this command, are set on the GUI of the custom print driver **1533** of the smartphone **500** irrespective of the sheet selection selected by the printer **300**. The page start command is a command to declare the start of a page image included in the print job. The print data transfer command is a command to transfer data, such as a bit image of the page image included in the print job. The page end command is a command to declare the end of the page image.

(150) A scan job command group includes a scan parameter setting command, a scan execution start command, a scan stop command, and so forth, but detailed description of these commands is omitted.

(151) A setting command group includes a registered sheet setting command, a setting reset command, and so forth.

(152) The registered sheet setting command is a command to set the sheet information data set **5005** included in the sheet information settings **5000**, which is sheet information that a user can register. This command includes the sheet size and the sheet type, selected by the user in the sheet information display section **604** of the sheet information-setting screen **600**, described with reference to FIG. **4A**, of the remote user interface displayed by the Web browser **1534** of the smartphone **500**. The setting reset command is a command to reset the settings of the printer **300**.

(153) A maintenance command group includes an error release command, a head cleaning execution command, an ink replacement execution command, a printer power-off command, and so forth.

(154) The error release command is a command to release an error occurring in the printer **300** to continue or stop the operation. The head cleaning execution command is a command to execute cleaning of a print head of the printer **300** so as to maintain the print quality. The ink replacement execution command is a command to execute a sequence for replacing an ink cartridge used in the printer **300**. The printer power-off command is a command to shift the printer **300** to the power-off state.

(155) FIG. **15** is a flowchart of the driver job execution process in the step **S2013** in FIG. **10**.

(156) In the present process, as shown in FIG. **15**, in a repeat loop of steps **S2401** to **S2407**, the commands included in the received driver job are sequentially read out, and processing according to each read command is executed. Note that FIG. **15** shows only processing operations performed in a case where out of the command group included in the received driver job, principal commands are read out, but description of processing operations performed in a case where the other commands included in the received driver job are read out is omitted.

(157) First, in a step **S2402**, a new command is read out from the received driver job.

(158) Next, in a step **S2403**, if the command read out in the step **2402** is the print data transfer command (YES to the step **S2403**), the print data included in this command is sequentially printed. More specifically, the following processing operations are executed: In a step **S2404**, the CPU **311** checks sheet matching (operation of a determination unit). To do this, the CPU **311** may check whether or not the sheet size and the sheet type, designated in the print parameter setting command which has already been read out from the driver job, match the sheet size and the sheet type of a data set indicated by a selected one of the sheet settings selection values **5001** (hereinafter referred to as “the set sheet settings selection value **5001**”). If there is a mismatch in at least one of the sheet size and the sheet type (NO to the step **S2404**), the CPU **311** (state shifting unit) shifts the printer **300** to an error state, and then proceeds to a step **S2405**, wherein the CPU **311** performs a sheet mismatch error-handling process. In the sheet mismatch error-handling process, the error is notified by using the error lamp **342** (notification unit) or the like, and confirmation is performed on whether to resolve the mismatch e.g., by a sheet selection operation or to print the data by setting the print job parameters while ignoring the mismatch. Further, the error notification may be transmitted to the client terminal **401** or the smartphone **500**, which is connected to the printer **300**. In this case, the client terminal **401** or the smartphone **500** having received the error notification may display information related to the error notification, such as details of the error and an error release method, on the screen. Note that in the present embodiment, details of the processing operations in the sheet mismatch error-handling process are differentiated based on whether the sheet size and the sheet type set for the received job correspond to one of the sheet selection lamps **333** to **335** or the sheet selection lamp **336**. If the sheet settings match each other (YES to the step **S2404**), or if the error state is released by executing the sheet mismatch error-handling process in the step **S2405**, the process proceeds to a step **S2406**, wherein the CPU **311** executes a print operation according to the bit image of the page image received by the print data transfer command.

(159) On the other hand, if the command read out in the step **S2402** is the print parameter setting command (NO to the step **S2403** and YES to a step **S2408**), the process proceeds to a step **S2409**. In the step **S2409**, the CPU **311** holds the parameters of the sheet size, the sheet type, and the print quality, designated in the command data, in the work area as the parameters for use in the print job. On the other hand, if the command read out in the step **S2402** is not the print parameter setting command (NO to the step **S2408**), the process proceeds to a step **S2410**.

(160) If it is determined in the step **S2410** that the command read out in the step **S2402** is the registered sheet setting command (operation of a detection unit) (YES to the step **S2410**), the process proceeds to a step **S2411**. On the other hand, if the command read out in the step **S2402** is not the registered sheet setting command (NO to the step **S2410**), the process proceeds to a step **S2412**.

(161) In the step **S2411**, the sheet size and the sheet type designated in the command data are stored in the sheet information data set **5005** of the sheet information settings **5000** (operation of a registration unit).

(162) If it is determined in the step **S2412** that the command read out in the step **S2402** is the job end command (YES to the step **S2412**), the process leaves the repeat loop and is terminated. On the other hand, if the command read out in the step **S2402** is not the job end command (NO to the step **S2412**), the process proceeds to the step **S2407** and returns to the step **S2402** of the repeat loop.

(163) FIG. **16** is a diagram showing a standard print job **5200** which is an example of a standard job generated by the standard print service **1522** of the smartphone **500**. Although in FIG. **16**, the standard print job **5200** whose job type is a print job will be described as an example of the standard job, the standard job may be of another job type, such as a scan job.

(164) In the printer **300**, if reception of a standard job from the smartphone **500** is detected in the step **S2014**, the received standard job is analyzed by the standard print application **1343**, and processing according to the analyzed contents of the job is executed.

(165) The standard print job **5200** is formed by a job header part **5201** and a job body part **5202**. The job header part **5201** is used to pass the parameters required to execute the standard print job **5200**.

(166) The job header part **5201** stores a job type **5203**, a job sender **5204**, a transmitting terminal **5205**, a transmission destination printer **5206**, a document name **5207**, a specified sheet size **5208**, a specified sheet type **5209**, print quality **5210**, and a specified number of copies **5211**.

(167) The job type **5203** stores the data of “print” indicating a print job which is a job type of the standard print job **5200**. Not that examples of the job type stored as the specification of the job type **5203** in the standard job include a print job and a scan job. The job sender **5204** stores a character string indicating a sender of the standard print job **5200**. The transmitting terminal **5205** stores a character string indicating a terminal which transmits the standard print job **5200**. The transmission destination printer **5206** stores a character string indicating the printer **300** that receives the standard print job **5200**. The document name **5207** stores a character string representing a document name of print data included in the standard print job **5200**. The specified sheet size **5208** stores a character string specifying a sheet size used in the standard print job **5200**. The character string specified here is different from the ID for identifying the sheet size in the printer **300**, and hence the printer **300** is required to perform conversion between the specified sheet size **5208** and the ID. The specified sheet type **5209** stores a character string specifying a sheet type used in the standard print job **5200**. The character string specified here is different from the ID for identifying the sheet type in the printer **300**, and hence the printer **300** is required to perform conversion between the specified sheet type **5209** and the ID. The print quality **5210** stores print quality required when the standard print job **5200** is executed. The specified number of copies **5211** stores the number-of-copies information used when the standard print job **5200** is executed.

(168) The job body part **5202** is used to pass the print data included in the standard print job **5200** to the printer **300**. Although FIG. **16** shows the contents of the job body part **5202** in a case where the job type **5203** is the print job, there is a case where the standard job is of another job type, a case where the job body part **5202** stores different contents, and a case where the job body part **5202** is not included.

(169) The job body part **5202** stores a number of pages **5212**, first page print image data **5213**, second page print image data **5214**, and so forth. Note that although FIG. **16** shows a case where the number of pages of the standard print job **5200** is two, by way of example, print image data is stored in the job body part **5202** for each page of the standard print job **5200**.

(170) The number of pages **5212** designates the number of image data items stored in the job body part **5202**, i.e., the number of pages of the standard print job **5200**. The first page print image data **5213** stores print image data of the first page included in the standard print job **5200**, and the second page print image data **5214** stores print image data of the second page included in the

standard print job **5200**. As the format of the print image data stored in the job body part **5202**, an image format which is generally used, such as TIFF and JPEG, or an image format having less image deterioration caused by compression, such as TIFF and PNG, can be used.

(171) FIG. **17** is a flowchart of the standard job execution process in the step **S2015** in FIG. **10**.

(172) First, in a step **S2501**, the CPU **311** identifies the job type of a standard job received from the smartphone **500** and executes processing operations according to the identified job type. For example, in a case where the received standard job is the standard print job **5200** shown in FIG. **16** by way of example. “print” is stored in the job type **5203**, and hence the job type is identified as a print job.

(173) In a step **S2502**, if the job type identified in the step **S2501** is the print job, the print data included in the received standard job is sequentially printed. More specifically, assuming that the received standard job is the standard print job **5200** (see FIG. **16**) (YES to the step **S2502**), the following processing operations are performed: In a step **S2503**, the CPU **311** checks sheet matching (operation of the determination unit). In this determination, the CPU **311** may check whether or not the sheet size and the sheet type, specified in the specified sheet size **5208** and the specified sheet type **5209** of the standard print job **5200**, match the sheet size and the sheet type of the data set associated with the set sheet settings selection value **5001**. If there is a mismatch (NO to the step **S2503**), the CPU **311** (state shifting unit) shifts the printer **300** to the error state, and then proceeds to a step **S2504** to perform the sheet mismatch error-handling process. In the sheet mismatch error-handling process, the CPU **311** notifies the user of the error using the error lamp **342** (notification unit) or the like, and confirmation is performed on whether to resolve the mismatch e.g., by a sheet selection operation or to print the data by setting the print job parameters while ignoring the mismatch. Further, the error notification may be transmitted to the client terminal **401** or the smartphone **500**, which is connected to the printer **300**. In this case, the client terminal **401** or the smartphone **500** having received the error notification may display information related to the error notification, such as details of the error and an error release method, on the screen. If there is no mismatch (YES to the step **S2503**), or if the error state is released by the sheet mismatch error-handling process in the step **S2504**, the process proceeds to a step **S2505**. In the step **S2505**, the CPU **311** acquires the first page print image data **5213** and the second page print image data **5214**, stored in the print job, and sequentially executes the print operation, followed by terminating the present process.

(174) On the other hand, if the job type identified in the step **S2501** is a scan job (NO to the step **S2502** and YES to a step **S2506**), the process proceeds to a step **S2507**, wherein the scan operation is executed, followed by terminating the present process. Note that the detailed description of the scan operation in this step is omitted. On the other hand, if the job type identified in the step **S2501** is not a scan job (NO to the step **S2506**), the process proceeds to a step **S2508**.

(175) In the step **S2508**, the CPU **311** executes processing according to the job type identified in the step **S2501**, followed by terminating the present process. Note that the detailed description of the processing in this step is omitted.

(176) According to the present embodiment, the user can change the sheet information data set **5005** (see FIG. **9**) included in the sheet information settings **5000** of the printer **300** from the sheet information-setting screen **600** (see FIG. **4A**) of the remote user interface. Further, with this change, the user can select not only the sheet settings already specified on the operation panel **305** of the printer **300**, but also the sheet settings corresponding to the sheet information data set **5005**. With this, even in a case where the display capability of the operation panel **305** of the printer **300** is limited, it is possible to use desired sheet settings.

(177) Note that in the present embodiment, the sheet information data set **5005** is set using the sheet information-setting screen **600** (see FIGS. **4A** to **4C**) of the remote user interface, by way of example. However, besides this, it is also possible to set the sheet information data set **5005** by receiving the registered sheet setting command (see FIG. **14**) included in the driver job from an

external terminal apparatus, such as the smartphone **500** or the client terminal **401**. For example, this can be realized by providing the same setting screen as the sheet information-setting screen **600**, shown in FIG. **4A**, on the GUI (setting screen) of the custom print driver **1533** (see FIG. **8**: printer driver) of the smartphone **500**. Further, this can be realized by separately installing a sheet settings registration application for the smartphone **500** for setting the sheet information data set **5005** and providing the same setting screen as the sheet information-setting screen **600**, shown in FIG. **4A**, on the GUI.

(178) Next, a second embodiment of the present invention will be described. In the sheet mismatch error-handling process in the first embodiment, the type of the sheet mismatch error to be notified is one type. In contrast, in the sheet mismatch error-handling process in the present embodiment, two types of the sheet mismatch error are notified. With this, the present embodiment enables the user to more easily grasp the cause of the sheet mismatch error than in the first embodiment, and compared with the case of the first embodiment, the user can more easily release the error state.

(179) The sheet mismatch error-handling process in the present embodiment will be described below with reference to FIG. **18**.

(180) FIG. **18** is a flowchart of the sheet mismatch error-handling process in the step **S2405** in FIG. **15** and the step **S2504** in FIG. **17** in the present embodiment.

(181) First, in a step **S2601**, the CPU **311** stores the sheet size ID of the job in a variable “a” in the work area. More specifically, when the driver job execution process (see FIG. **15**) is being executed, the sheet size ID included in the print parameter setting command of the driver job is stored in the variable “a”. On the other hand, when the standard job execution process (see FIG. **17**) is being executed, the specified sheet size included in the job header part of the standard job is converted to the corresponding sheet size ID and stored in the variable “a”.

(182) In the next step **S2602**, the CPU **311** stores the sheet type ID of the job in a variable “b” in the work area. Details of this processing is the same as the procedure for acquiring the sheet size ID in the step **S2601**, and hence detailed description thereof is omitted.

(183) In the next step **S2603**, a value of 1 is stored in a variable “c” as its initial value in the work area holding one of the IDs of the sheet information data sets **5002** to **5005**. That is, the initial value of the variable “c” indicates the sheet information data set **5002** in the above-described sheet information settings **5000** shown in FIG. **9**.

(184) In the next step **S2604**, as the initial value of the number of repetitions, a value of 1 is stored in a variable “n” in the work area.

(185) Then, repeated processing of steps **S2605** to **S2609** is executed. In this repeated processing, first, in a step **S2606**, the sheet size ID and the sheet type ID, included in the sheet information data set indicated by the variable “c”, are acquired. Then, the CPU **311** checks whether or not the acquired sheet size ID and the variable “a” match each other and whether or not the acquired sheet type ID and the variable “b” match each other. If the checks both indicate a match between the acquired ID and the variable, the process leaves the repeated processing and the process proceeds to a step **S2610**, whereas if at least one of the checks indicates a mismatch, the process proceeds to a step **S2607**, wherein the CPU **311** increments the variable “c” indicating the sheet information data set by 1. Then, in a step **S2608**, the CPU **311** increments the variable “n” indicating the number of repetitions by 1. Then, in a step **S2609**, if the variable “n” indicating the number of repetitions is equal to or larger than 5, the process leaves the repeated processing, and proceeds to the step **S2610**, whereas if the variable “n” is smaller than 5, the process returns to the step **S2605** to continue the repeated processing.

(186) In the step **S2610**, the CPU **311** checks whether or not the variable “n” indicating the number of repetitions is smaller than 5. Note that this determination corresponds to the determination of whether or not the sheet settings (the sheet size and the sheet type) designated in the job is included in the plurality of sheet settings sets which can be currently set by an operation performed on the main unit of the printer **300** (operation performed on the sheet selection key **337**). This is because

in a case where a sheet settings set designated in the job is not included in the plurality of sheet settings sets which can be currently set by an operation performed on the main unit of the printer **300**, the variable “n” becomes equal to or larger than 5. That is, a case where the answer to this determination becomes negative (NO) is a case where the sheet settings set designated in the job can be set on the remote user interface as the sheet settings set corresponding to the sheet selection lamp **336** but is not currently set as the sheet settings set associated with the sheet selection lamp **336**. If the variable “n” is smaller than 5 (YES to the step **S2610**), the printer **300** is shifted to the error state, and then the process proceeds to a step **S2611**. In the step **S2611**, the CPU **311** notifies the user of a first sheet mismatch error (first notification), whereby release of the error is awaited by prompting the user to operate the sheet selection key **337** included in the operation panel **305**. More specifically, the CPU **311** notifies the user that the first sheet mismatch error has occurred by blinking the error lamp **342** included in the operation panel **305**. The first sheet mismatch error is an error caused by a reason that the sheet settings set designated in the job has not been currently set for the printer **300** by the sheet selection key **337** but can be set by operating the sheet selection key **337**. Further, the CPU **311** notifies the client terminal **401** or the smartphone **500**, which is connected to the printer **300**, that the first sheet mismatch error has occurred. Note that the client terminal **401** or the smartphone **500** having received this notification may display information based on the received notification on a notification screen for notifying the user of the contents to be notified to the user, including information that the first sheet mismatch error has occurred, details of the error, and a method of releasing the error. FIG. **19A** shows an example of the notification screen displayed at this time. For example, the screen displayed based on the notification that the first sheet mismatch error has occurred includes an area for notifying the user that the print settings concerning the sheet designated by the job and the sheet settings currently set for the printer **300** do not match. Note that the notification screen may be a screen that displays a link button for displaying the above-described contents to be notified to the user using the Web browser. In the illustrated example in FIG. **19A**, when a link button **19A1** of “instruction manual” is pressed, a Web page indicating the method of releasing the error is displayed by the Web browser. Further, when a “print stop” button **19A2** is pressed, an instruction for canceling the job is transmitted from the client terminal **401** or the smartphone **500**, which has received pressing of this button, to the printer **300**. When the printer **300** receives the job cancellation instruction, even when the sheet mismatch error-handling process in FIG. **18** is being executed, cancellation of the job may be executed, and the sheet mismatch error-handling process in FIG. **18** may be terminated. In the next step **S2612**, the CPU **311** starts the lamp blinking operation for blinking one of the sheet selection lamps **333** to **336**, which corresponds to the sheet information data set indicated by the variable “c”. The variable “c” at this time is a value compared with and determined to match the variable “a” indicating sheet size ID of the job, in the step **S2606**, and as a result, the lamp corresponding to the sheet information data set indicated by the variable “c” is a lamp corresponding to the sheet settings designated by the job. Note that this lamp blinking operation is executed as another process operating on the real-time OS and executed in parallel with the present process. Note that the blinking pattern of the lamp in the lamp blinking processing may be the same as or different from the blinking pattern of the lamp in the lamp blinking operation started in the step **S2202** in FIG. **12**.

(187) In the next step **S2613**, the CPU **311** turns on one of the sheet selection lamps **333** to **336**, which corresponds to the set sheet settings selection value **5001** of the sheet information settings **5000**. The lamp corresponding to the set sheet settings selection value **5001** of the sheet information settings **5000** is the lamp corresponding to the sheet settings currently set for the printer **300**. In the next step **S2614**, the CPU **311** waits for occurrence of an event, and when a predetermined event occurs, the process proceeds to a step **S2615**. The predetermined event whose occurrence is monitored for in this step is a key operation performed by the user on the operation panel **305**.

(188) If it is detected in the step S2615 that the occurred event is the operation of pressing the sheet selection key **337** (YES to the step S2615), the process proceeds to a step S2616, wherein the sheet selection operation process described with reference to FIG. 12 is executed. At this time, although the lamp turned on in the step S2613 is turned off, two lamps out of the sheet selection lamps **333** to **336** are in a blinking state as a result of execution of the step S2612 and the step S2202 in FIG. 12. The user presses the sheet selection key **337** while confirming the state of the sheet selection lamps **333** to **336**, and when the number of blinking lamps is reduced to one, the user presses the OK key **338** to determine the sheet settings selection value **5001**. With this, it is possible to cause the sheet settings of the sheet information data set having the ID set to the sheet settings selection value **5001** to match the sheet settings of the job. When the sheet selection operation process is terminated, the process proceeds to a step S2617, wherein the CPU **311** checks sheet matching. In this check, the CPU **311** is only required to check whether or not the sheet size ID and the sheet type ID, which are stored as the variable “a” and the variable “b”, respectively, match the sheet size ID and the sheet type ID of the sheet information data set having the ID set to the sheet settings selection value **5001**. If it is determined as a result of the check that a sheet mismatch has occurred (NO to the step S2617), the process returns to the step S2613, whereas if a sheet match has occurred (YES to the step S2617), the process proceeds to a step S2618, wherein the sheet mismatch error is released, followed by terminating the present process. In this step, the sheet mismatch error is released by turning off the error lamp **342** included in the operation panel **305**. Further, the one of the sheet selection lamps **333** to **336**, which corresponds to the sheet settings selection value **5001** of the sheet information settings **5000**, is turned on, and the other lamps are turned off.

(189) On the other hand, if it is detected that the occurred event is the operation of pressing the OK key **338** (NO to the step S2615 and YES to a step S2619), the process proceeds to the step S2618, wherein the sheet mismatch error is released, followed by terminating the present process. In this case, the print job parameters are set by ignoring the mismatch.

(190) Note that if it is detected that the occurred event is neither the operation of pressing the sheet selection key **337** nor the operation of pressing the OK key **338** (NO to the step S2615 and NO to the step S2619), the process returns to the step S2614.

(191) If it is determined in the step S2610 that the variable “n” is equal to or larger than 5 (NO to the step S2610), the process proceeds to a step S2620, wherein the CPU **311** notifies the user of a second sheet mismatch error (second notification) and prompts the user to perform the sheet information setting on the remote user interface to release the error without prompting the user to perform the sheet selection operation. The second sheet mismatch error is an error caused by a reason that the sheet settings designated in the job is not currently set for the printer **300** by the sheet selection key **337** and at the same time is not included in the plurality of sheet settings which can be set by operating the sheet selection key **337**. The plurality of sheet settings which can be set by operating the sheet selection key **337** are the sheet settings which are currently associated with the sheet selection lamps **333** to **336**. More specifically, the following operation is performed: In the step S2620, occurrence of the second sheet mismatch error is notified by blinking the error lamp **342** included in the operation panel **305**. Note that the error lamp **342** may be blinked according to a blinking pattern different from the blinking pattern of the error lamp **342** in the step S2611.

(192) Further, the CPU **311** notifies the client terminal **401** or the smartphone **500**, which is connected to the printer **300**, that the second sheet mismatch error has occurred. Note that the client terminal **401** or the smartphone **500** having received this notification may display information based on the received notification on a notification screen for notifying contents to be notified to the user, including information that the second sheet mismatch error has occurred, details of the error, and a method of releasing the error. FIG. 19B shows an example of the notification screen displayed at this time. In the present embodiment, the screen displayed based on the notification

notifying the user of occurrence of the first sheet mismatch error and the screen displayed based on the notification notifying the user of occurrence of the second sheet mismatch error are differentiated. For example, the screen displayed based on the notification that the second sheet mismatch error has occurred includes an area for notifying the user that the print settings concerning the sheet designated by the job is not included in the plurality of sheet settings which can be currently set by performing an operation on the body of the printer **300**.

(193) Further, for example, the screen includes an area for prompting the user to cancel the job by pressing the stop key **341**. Further, for example, the screen includes an area for notifying the user that the settings concerning the sheet, set for the printer **300**, can be changed on the remote user interface. Note that the notification screen may be a screen that displays a link button for displaying the above-described contents to be notified to the user using the Web browser. In the illustrated example in FIG. **19B**, when a link button **19B1** of “instruction manual” is pressed, a Web page indicating the method of releasing the error is displayed by the Web browser. Further, when a “print stop” button **19B2** is pressed, an instruction for canceling the job is transmitted from the client terminal **401** or the smartphone **500**, which has received pressing of this button, to the printer **300**. When the printer **300** receives the job cancellation instruction, even when the process in FIG. **18** is being executed, cancellation of the job may be executed, and the process in FIG. **18** may be terminated.

(194) Further, in a case where the second sheet mismatch error has occurred, the printer **300** may perform control to invalidate an operation from the sheet selection key **337** to thereby inhibit the current sheet settings of the printer **300** from being changed by operating the sheet selection key **337**. This is because the second sheet mismatch error is not solved by changing the current sheet settings of the printer **300** by operating the sheet selection key **337**.

(195) In the next step **S2621**, the CPU **311** turns off all of the sheet selection lamps **333** to **336** to indicate the sheet size ID and the sheet type ID, stored in the variable “a” and the variable “b”, respectively, do not match any of the sheet information data sets included in the sheet information settings **5000**. Note that in the step **S2621**, not all of the sheet selection lamps **333** to **336** are turned off, but the same processing as the step **S2613** may be executed.

(196) In the next step **S2622**, the CPU **311** waits for occurrence of an event, and when a predetermined event occurs, the process proceeds to a step **S2623**. The predetermined event whose occurrence is monitored for in this step is one of a key operation performed by the user on the operation panel **305** and a remote user interface access request from the smartphone **500**.

(197) If it is detected in the step **S2623** that the occurred event is the operation of pressing the OK key **338** (YES to the step **S2623**), the process proceeds to the step **S2618**, wherein the sheet mismatch error is released, followed by terminating the present process.

(198) On the other hand, if it is detected that the occurred event is not the operation of pressing the OK key **338** (NO to the step **S2623**), the process proceeds to a step **S2624**. Then, in the step **S2624**, it is determined whether or not the occurred event is a remote user interface access request from the smartphone **500**. If it is detected that the occurred event is a remote user interface access request from the smartphone **500** (NO to the step **S2623** and YES to the step **S2624**), the process proceeds to a step **S2625**. In the step **S2625**, the sheet information-setting process from the remote user interface, described with reference to FIG. **13**, is executed to once release the sheet mismatch error. With this sheet information-setting process, the registered sheet information is updated, whereby it is possible to cause the sheet size ID and the sheet type ID, stored in the variable “a” and the variable “b”, respectively, to match the sheet information data set **5005** included in the sheet information settings **5000**. When the sheet information-setting process is terminated, the step **S2603** et seq. are executed. With this, the CPU **311** checks whether or not the sheet size ID and the sheet type ID, stored in the variable “a” and the variable “b”, respectively, and the sheet size ID and the sheet type ID included in one of the four sheet information data sets **5002** to **5005** included in the sheet information settings **5000** match.

(199) Note that in a case where a user operation of pressing the stop key **341** is received when the CPU **311** waits for occurrence of an event in the step **S2614** and the **S2622**, cancellation of the job may be executed, and the present process may be terminated. However, in a case where the present process (in the step **S2405**) is terminated by executing cancellation of the job during the driver job execution process (see FIG. **15**), a step for determining whether or not cancellation of the job has been executed is provided immediately after the step **S2405**. If it is determined in this step that cancellation of the job has been executed, the CPU **311** is only required to perform control not to execute the step **S2406**. Similarly, in a case where the present process (in the step **S2504**) is terminated by executing cancellation of the job during the standard job execution process (see FIG. **17**), a step for determining whether or not cancellation of the job has been executed is provided immediately after the step **S2504**. If it is determined in this step that cancellation of the job has been executed, the CPU **311** is only required to perform control not to execute the step **S2505**. Further, although in the above description, the sheet mismatch error is resolved by executing the sheet information-setting process from the remote user interface before receiving cancellation of the job, this is not limitative. Execution of the sheet information-setting process from the remote user interface may be controlled not to be executed unless cancellation of the job is received. That is, unless cancellation of the job is received, even when a remote user interface access request is received from the smartphone **500**, the CPU **311** may perform control to ignore this request and perform control to prevent the sheet information-setting process from being executed from the remote user interface.

(200) Note that when the sheet mismatch error-handling process in FIG. **18** is terminated in the step **S2405** of the driver job execution process in FIG. **15**, printing is executed in the step **S2406**. Incidentally, if the answer to the question of the step **S2615** is negative (NO) and the answer to the question of the step **S2619** is affirmative (YES) at the same time, the error is released in the step **S2618**, but in actuality, the sheet settings in the job and the sheet settings currently set for the printer **300** remain unmatched. In this case, in the printing executed in the step **S2406**, it is assumed that an image of a size based on the sheet settings of the job is printed on a sheet based on the sheet settings currently set for the printer **300**. However, this is not limitative. If the answer to the question of the step **S2615** is negative (NO) and the answer to the question of the step **S2619** is affirmative (YES) at the same time, it may be further determined whether or not the size of the sheet based on the sheet settings currently set for the printer **300** is larger than the size based on the sheet settings of the job. Then, if the answer to this question is affirmative (YES), the process may proceed to the step **S2618**, whereas if the answer to this question is negative (NO), the error may be notified without executing printing based on the job while terminating the sheet mismatch error-handling process in FIG. **18**, and also the CPU **311** may wait for a change of the sheet settings currently set for the printer **300**. Then, when the current sheet settings of the printer **300** are changed, the process may return to the step **S2503**. With this, it is possible to prevent an image from being printed such that the image extends off a sheet based on the sheet settings currently set for the printer **300**.

(201) As described above, in the present embodiment, in a case where a sheet mismatch error has occurred, whether the sheet mismatch error having occurred is the first sheet mismatch error or the second sheet mismatch error is determined. Note that this determination corresponds to the steps **S2605** to **S2610**. Then, in the present embodiment, different processing is executed according to whether the sheet mismatch error having occurred is the first sheet mismatch error or the second sheet mismatch error. In other words, processing executed based on occurrence of the first sheet mismatch error and processing executed based on occurrence of the second sheet mismatch error are controlled to be differentiated. More specifically, in the present embodiment, the processing executed to resolve (release) the sheet mismatch error having occurred is different according to whether the sheet mismatch error having occurred is the first sheet mismatch error or the second sheet mismatch error. The processing executed to resolve the first sheet mismatch error having

occurred corresponds to the steps **S2614** to **S2617** and **S2619**, and the processing executed to resolve the second sheet mismatch error having occurred corresponds to the steps **S2622** to **S2625**. Further, the contents of the notification processing for notifying the user of occurrence of the sheet mismatch error is different according to whether the sheet mismatch error having occurred is the first sheet mismatch error or the second sheet mismatch error. The notification processing associated with the first sheet mismatch error corresponds to the steps **2611** to **S2613**, and the notification processing associated with the second sheet mismatch error corresponds to the steps **2620** and **S2621**.

(202) With the above-described control, in the printer **300** that enables a user to use desired sheet settings, whether to prompt the user to operate the sheet selection key **337** included in the operation panel **305** or prompt the user to set sheet information settings on the remote user interface is determined. With this, it is possible to provide the user with a different notification of the sheet mismatch error according to a result of the determination. As a result, even in a case where there is a limit in the display capability of the operation panel **305** of the printer **300**, the user can execute the operation for releasing the sheet mismatch error, and it is possible to improve the operability of the printer **300**.

(203) Note that in the present embodiment, the present invention can also be realized by supplying a program that realizes one or more functions to a system or a computer of an apparatus via a network or a recording medium, and the system or a system controller of the apparatus loading and executing the program. The system controller has one or a plurality of processors or circuits and can include a network of a plurality of separate system controllers or a plurality of separate processors or circuits so as to load and execute executable commands.

(204) The processors or circuits can include a central processing unit (CPU), a micro processing unit (MPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), and a field programmable gate array (FPGA). Further, the processors or circuits can include a digital signal processor (DSP), a data flow processor (DFP), or a neural processing unit (NPU).

OTHER EMBODIMENTS

(205) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(206) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A printing apparatus to which is applied a sheet setting selected by a user out of a plurality of sheet settings including a plurality of first sheet settings and a second sheet setting, wherein the first sheet settings are settings for which pieces of first sheet information on first sheets determined in advance without a user's selection and are registered in the printing apparatus, wherein the second sheet setting is a setting for which second sheet information on a second sheet is registered in the printing apparatus, the second sheet being selected by the user out of a plurality of sheets on a remote user interface displayed by a web browser function of an external apparatus communicable with the printing apparatus, wherein sheet information registered for the second sheet setting in the printing apparatus is changeable to other sheet information via the remote user interface, the printing apparatus comprising: a receiver configured to receive a print job; a printer configured to execute printing based on the print job; and a controller having a memory containing instructions and a processor for executing the instructions, and configured to in a case where a user operation to register the second sheet information in the printing apparatus is performed on the remote user interface, register the second sheet information in the printing apparatus for the second sheet setting, execute a first process before the printing based on the received print job is executed, based on occurrence of a first error that sheet information set in the received print job is different from sheet information registered in the printing apparatus for a sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and also the sheet information set in the received print job is neither any of the pieces of first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface, and execute a second process different from the first process before the printing based on the received print job is executed, based on occurrence of a second error that the sheet information set in the received print job is different from the sheet information registered in the printing apparatus for the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, but the sheet information set in the received print job is any of the pieces of first sheet information or the second sheet information registered in the printing apparatus via the remote user interface, wherein the first process includes, in a case where the sheet information set in the received print job is different from the sheet information registered for the sheet setting applied to the printing apparatus when the print job has been received and where the sheet information set in the received print job is neither any of the pieces of first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface, displaying on the remote user interface, a first notification screen that prompts the user to change, on the remote user interface, the second sheet information registered in the printing apparatus via the remote user interface, and wherein the second process includes displaying on the remote user interface a second notification screen, different from the first notification screen, that prompts the user to confirm the sheet setting applied to the printing apparatus.
2. The printing apparatus according to claim 1, wherein the printing based on the print job is executed without executing the first process and the second process based on a match between the sheet information set in the received print job and the sheet information associated with the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received.
3. The printing apparatus according to claim 1, wherein the first notification screen is displayed on a display of the external apparatus when the first notification is executed on the external apparatus, and wherein the second notification screen is displayed on the display of the external apparatus when the second notification is executed.

4. The printing apparatus according to claim 3, wherein a predetermined button for canceling execution of the received print job is included in the first notification screen and the second notification screen, wherein in a case where the predetermined button is operated, an instruction for canceling execution of the received print job is transmitted to the printing apparatus, and wherein in a case where the instruction is received by the printing apparatus, execution of the received print job is cancelled.
5. The printing apparatus according to claim 3, wherein the first notification screen is a screen for displaying a method of releasing the first error, and wherein the second notification screen is a screen for displaying a method of releasing the second error.
6. The printing apparatus according to claim 3, wherein the second notification screen includes an area for a notification related to a remote setting function for selecting sheet information to be associated with the second sheet setting on the external apparatus.
7. The printing apparatus according to claim 1, further comprising: a controller configured to receive a selection operation for selecting one of the plurality of sheet settings from a user, and a changing controller configured to change a sheet setting applied to the printing apparatus, based on the selection operation, and wherein the second process includes processing for performing control to prevent the sheet setting applied to the printing apparatus from being changed by the selection operation.
8. The printing apparatus according to claim 7, wherein the processing for performing control to prevent the sheet setting applied to the printing apparatus from being changed by the selection operation is processing for invalidating the selection operation.
9. The printing apparatus according to claim 7, wherein the first process includes processing for detecting the selection operation, and wherein in a case where the selection operation is detected, the sheet setting applied to the printing apparatus is changed based on the selection operation.
10. The printing apparatus according to claim 1, further comprising: a controller configured to receive a selection operation for selecting one of the plurality of sheet settings from a user, and a changing controller configured to change the sheet setting applied to the printing apparatus based on the selection operation, and wherein in a case where the second error has occurred, printing is executed based on the print job after the sheet setting applied to the printing apparatus is changed based on the selection operation.
11. The printing apparatus according to claim 10, wherein in a case where the second error has occurred, and where the sheet setting applied to the printing apparatus has been changed to the first sheet setting based on the selection operation, printing is executed based on the print job and the first sheet setting.
12. The printing apparatus according to claim 1, further comprising a second receiver configured to receive an operation of pressing a predetermined button from a user, wherein in a case where the first error or the second error has occurred, where the predetermined button has been pressed without changing the sheet setting applied to the printing apparatus based on the selection operation, and where a size corresponding to the sheet information associated with the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, is larger than a size corresponding to the sheet information set in the received print job, printing based on the print job is executed without changing the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and wherein in a case where the first error or the second error has occurred, where the predetermined button has been pressed without changing the sheet setting applied to the printing apparatus based on the selection operation, and where a size corresponding to the sheet information associated with the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received is not larger than a size corresponding to the sheet information set in the received print job, control not to execute printing based on the print job is performed.
13. The printing apparatus according to claim 1, wherein the second process includes processing

for detecting an access request from a remote setting function for selecting sheet information to be associated with the second sheet setting on an external apparatus communicable with the printing apparatus, wherein in a case where the access request is detected, the sheet information associated with the second sheet setting is changed based on the remote setting function, and wherein after the sheet information associated with the second sheet setting has been changed based on the remote setting function, printing is executed based on the print job and the second sheet setting.

14. The printing apparatus according to claim 1, wherein the printing apparatus includes a plurality of lamps, wherein the first process includes processing for controlling the plurality of lamps to a first state, and wherein the second process includes processing for controlling the plurality of lamps to a second state different from the first state.

15. The printing apparatus according to claim 14, wherein the first state includes at least one of a state in which a blinking lamp corresponds, out of the plurality of lamps, to a sheet setting with which is associated a sheet setting matching the sheet setting set in the received print job, a state in which a blinking lamp corresponds, out of the plurality of lamps, to the sheet setting applied to the printing apparatus, and a state in which an error lamp, out of the plurality of lamps, which is different from a plurality of lamps corresponding to the plurality of sheet settings is turned on blinks.

16. The printing apparatus according to claim 14, wherein the second state includes at least one of a state in which a plurality of lamps are turned off which correspond, out of the plurality of lamps, to the plurality of sheet settings, and a state in which an error lamp, out of the plurality of lamps, which is different from the plurality of lamps corresponding to the plurality of sheet settings is turned on or blinks.

17. The printing apparatus according to claim 14, wherein the plurality of sheet settings further include a third sheet setting, and wherein the third sheet settings is a setting associated with third sheet information determined in advance without the user's selection.

18. The printing apparatus according to claim 1, wherein the first sheet is at least one of an A4 sheet, a letter sheet, and a 4×6 sheet.

19. The printing apparatus according to claim 1, wherein the plurality of sheets include at least one of an A4 sheet, an A5 sheet, an A6 sheet, a B5 sheet, a B6 sheet, a letter sheet, a 4×6 sheet, a postcard, and an envelope.

20. The printing apparatus according to claim 1, wherein the sheet information includes at least one of a sheet size and a sheet type.

21. The printing apparatus according to claim 1, further comprising a plurality of lamps, wherein the second process includes executing the second notification by using the second notification screen that prompts, due to the difference, the user to confirm a status of a lamp out of the plurality of lamps, corresponding to the sheet setting applied to the printing apparatus.

22. The printing apparatus according to claim 1, wherein the first notification screen includes information indicating that the sheet information set in the received print job is not applicable to the printing apparatus.

23. The printing apparatus according to claim 1, wherein the first notification screen includes information that prompts the user to press a print stop button of the printing apparatus to stop the printing and restart the printing after changing a sheet setting.

24. The printing apparatus according to claim 1, wherein the pieces of first sheet information include A4 size, letter size, and 4×6 size.

25. A method of controlling a printing apparatus to which is applied a sheet setting selected by a user out of a plurality of sheet settings including a plurality of first sheet settings and a second sheet setting, wherein the first sheet settings are settings for which pieces of sheet information on first sheets determined in advance without a user's selection and are registered in the printing apparatus, wherein the second sheet setting is a setting for which sheet information on a second sheet is registered in the printing apparatus, the second sheet being selected by the user out of a

plurality of sheets on a remote user interface displayed by a web browser function of an external apparatus communicable with the printing apparatus, wherein sheet information registered for the second sheet setting in the printing apparatus is changeable to other sheet information via the remote user interface, the method comprising: receiving a print job; in a case where a user operation to register the second sheet information in the printing apparatus is performed on the remote user interface, registering the second sheet information in the printing apparatus for the second sheet setting; executing printing based on the print job; executing a first process before the printing based on the received print job is executed, based on occurrence of a first error that sheet information set in the received print job is different from sheet information registered in the printing apparatus for a sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and also the sheet information set in the received print job is neither any of the pieces of first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface; and executing a second process different from the first process before the printing based on the received print job is executed, based on occurrence of a second error that the sheet information set in the received print job is different from the sheet information registered in the printing apparatus for the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, but the sheet information set in the received print job is any of the pieces of first sheet information or the second sheet information registered in the printing apparatus via the remote user interface, in executing the first process, in a case where the sheet information set in the received print job is different from the sheet information registered for the sheet setting applied to the printing apparatus when the print job has been received and where the sheet information set in the received print job is neither any of the pieces of first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface, displaying on the remote user interface a first notification screen that, prompts the user to change, on the remote user interface, the second sheet information registered in the printing apparatus via the remote user interface, and in executing the second process displaying on the remote user interface a second notification screen, different from the first notification screen, that prompts the user to confirm the sheet setting applied to the printing apparatus.

26. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a method of controlling a printing apparatus to which is applied a sheet setting selected by a user out of a plurality of sheet settings including a plurality of first sheet settings and a second sheet setting, wherein the first sheet settings are settings for which pieces of sheet information on first sheets determined in advance without a user's selection and are registered in the printing apparatus, wherein the second sheet setting is a setting for which second sheet information on a second sheet is registered in the printing apparatus, the second sheet being selected by the user out of a plurality of sheets on a remote user interface displayed by a web browser function of an external apparatus communicable with the printing apparatus, wherein sheet information registered for the second sheet setting in the printing apparatus is changeable to other sheet information via the remote user interface, wherein the method comprises: receiving a print job; in a case where a user operation to register the second sheet information in the printing apparatus is performed on the remote user interface, registering the second sheet information in the printing apparatus for the second sheet setting; executing printing based on the print job; executing a first process before the printing based on the received print job is executed, based on occurrence of a first error that sheet information set in the received print job is different from sheet information registered in the printing apparatus for a sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, and also the sheet information set in the received print job is neither any of the pieces of first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface; and executing a second process different from the first process before the printing based on the received print job is executed,

based on occurrence of a second error that the sheet information set in the received print job is different from the sheet information registered in the printing apparatus for the sheet setting applied, out of the plurality of sheet settings, to the printing apparatus when the print job has been received, but the sheet information set in the received print job is any of the pieces of first sheet information or the second sheet information registered in the printing apparatus via the remote user interface, in executing the first process, in a case where the sheet information set in the received print job is different from the sheet information registered for the sheet setting applied to the printing apparatus when the print job has been received and where the sheet information set in the received print job is neither any of the pieces first sheet information nor the second sheet information registered in the printing apparatus via the remote user interface, displaying on the remote user interface a first notification screen that prompts the user to change, on the remote user interface, the second sheet information registered in the printing apparatus via the remote user interface, and in executing the second process displaying on the remote interface a second notification screen, different from the first notification screen, that prompts the user to confirm the sheet setting applied to the printing apparatus.
